

Alex Rivera

Computer Science student - Delft University of Technology

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Summary

Third-year Computer Science student who builds backend services, data pipelines and the tooling around them. Looking for a summer 2027 software engineering internship on a backend or infrastructure team.

Education

B.Sc. Computer Science

Delft University of Technology | 2023 - expected 2027

- Coursework: Algorithms & Data Structures, Computer Networks, Databases, Operating Systems, Distributed Systems, Compiler Construction, Machine Learning.

Experience

Software Engineering Intern

Fathom Labs (student startup, 12 people) | Jul - Sep 2025

- Moved the reporting service off a nightly cron job onto an incremental pipeline, cutting the morning job from 22 minutes to under 4.
- Added integration tests around the ingestion path that had broken twice that summer.

Teaching Assistant, Algorithms & Data Structures

Delft University of Technology | Feb 2025 - present

- Run weekly labs for 40 students and grade 200+ assignments a semester.
 - Rewrote the graph-algorithms lab after three cohorts in a row hit the same edge case in Dijkstra's algorithm.
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Selected projects

CourseGraph - TypeScript, Next.js, PostgreSQL, D3

github.com/alexrivera/coursegraph

- Prerequisite map for 1,200+ catalogue courses; a topological sort over the parsed data surfaced 14 circular prerequisite chains nobody had noticed.

Seatwatch - Python, SQLite, Docker, GitHub Actions

github.com/alexrivera/seatwatch

- Course-registration notifier used by around 400 students; a token-bucket queue with jittered backoff keeps it inside the registration API's rate limit.

raft-kv - Go, gRPC, Protocol Buffers

github.com/alexrivera/raft-kv

- Fault-tolerant key-value store implementing leader election, log replication and snapshotting, tested against a simulator that drops, delays and partitions messages.
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Skills

Languages: TypeScript, Python, Go, SQL, Rust, C

Frameworks: Next.js, React, Node.js, FastAPI, gRPC, D3

Data & infrastructure: PostgreSQL, Redis, SQLite, Docker, GitHub Actions, Linux